

Monsoon Semester, 2023, Faculty of Architecture, CEPT University

Code	AR3052	Course	Resilience Thrift: Strategies for Adapting to Climatic Change in Housing for the Urban Poor		
Typology	Studio	Credits	14	GPA/NonGPA	GPA

Long Description :

The consequences of climate change – droughts, rising temperatures, increased flood events, water crisis, depleted groundwater levels and many more – have affected millions of people in India. On top of this, urbanization increases disasters' severity by adding population to cities whose infrastructure services are already limited. The urban poor, settling in the areas lacking infrastructure services, are the most vulnerable to climate-induced disasters and events. Many of Ahmedabad's low-income settlements are at a high risk of flooding, facing shortages of freshwater supplies and other essential resources. Informal settlements are densely populated, poorly ventilated, and often made from corrugated metal and plastic sheets, which cannot defend against the vagaries of natural calamities or increasingly adverse heat conditions. The intersection of poor construction quality with settlement locations concentrated on sites of elevated exposure to climate disasters drastically increases risks to these populations. Climate-sensitive diseases like diarrhoea, dengue fever, chikungunya and malaria, dehydration, heat cramps, heat exhaustion, and heat stroke are on a rise in such areas. This studio works together with SAATH to co-create local solutions to help these communities adapt and build their own resilience through design research and built pilot projects. The FA studio will be taught as a pair with a BUD studio. In the architectural studio we will concentrate in the first 12 weeks on analysing and engaging at an architectural scale to understand how exposure to climatic risk is not distributed evenly across buildings in informal settlements and to design modifications that mitigate conditions in the most extreme cases. We will begin by studying groups of houses to identify those least equipped to cope with climatic events and conditions. The analysis will include heat measurements as well as an architectural evaluation. We will then be selecting individual buildings as sites for which students will design retrofits to enhance passive climatic resilience. These will be discussed with the residents within the community, developed or modified, and then prototyped (at cept) to measure their effectiveness. The data from such testing can support future implementation on site. We will be engaging with the urban design studio at various points in this process to understand how socio-political factors and urban level analysis and architectural scale responses can inform each other. In weeks 12 - 16 we will be designing new buildings incorporating principles, strategies and devices from the first part of the semester. These are yet to be determined but may include new build housing solutions or community level infrastructural proposals emerging from the work of the BUD students in the first part of the semester.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Develop and manage the execution of residential weather resilient building elements.
- Measure and interpret the climatic performance of designed building elements.
- Value engineer and estimate the cost of designed building elements.
- Conduct and interpret structured and semi structured community interviews and discussions.
- Critically analyse the relevance and potential of building retrofits as a climatic strategy for informal housing.

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	Contextualisation and research
Week 2	Architectural scale sites
Week 3	Architectural scale sites
Week 4	Individual Building Documentation
Week 5	Design (Proposal to mitigate risk in individual selected building)
Week 6	Design
Week 7	Design
Week 8	Design
Week 9	Community Consultation
Week 10	Design Modifications
Week 11	Making of Selected Devices
Week 12	Testing and Measurement
Week 13	Speculative Design Brief
Week 14	Design
Week 15	Design
Week 16	Final Review

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Code	AR3604	Course	HRW-5 Perspectives on the Architecture of the Global South		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

Occidental forces have shaped architectural thinking to the extent that it is difficult to discuss architecture outside this framework. Yet, geographically, a large part of the world that includes Australasia, Africa and South America exists outside this narrative. It is marginalised in most discourses, with the exception of projects undertaken by architects from Europe, UK or North America, or for its exoticness. In more recent times, a diversity of 'others' have sought centre stage, and the 'global south' is no exception to this development. Marginalisation, on the basis of geography, economy, gender and community form the basis of the definition of global south. These factors are interlinked, and often not separated from each other. For example, colonisation of a large part of Asia, Africa and South America contributed to their marginalisation from the development narrative, in particular industrialisation. Even as late as the 20th century terms like 'first world' and third world were common place use in international organisations including the UN. The emergence of the 'global south' can be attributed to the growth and development seen in regions of the world other than Europe, US and UK. The post-colonial political restructuring postulated a new world order, and the tendency to organize the world as us and them became stronger. In this context, architecture as a representation of the development of society, becomes a significant tool and expression of the world order. However, there are very few texts that discuss the phenomenon of the global south and its impact on architecture. This course introduces students to the different trajectories of architectural practice in the Global South. Analysis and interpretation provide basis for perspectives on architecture not defined by European frameworks. The course rests on the premise that the 'north' and the 'south' are both parts of the same world, and their differentiation is constructed through global forces that have their own historiography. Practices and forces that have shaped the architecture in the 'other' part of the world will be discussed through case studies of a body of works placed in the decades from 1960s to 2020. The lectures will cover a series of topics starting from the decoding of the terminologies that constructed the 'global south' to defining themes and issues that form the broad field of architecture. The course is organised geographically. The course is divided into three parts. The first part presents an overview of the architectural practices in the global south using geographical proximities as a basis. The second part will extract thematic similarities across the region to articulate the position of the various architects on dominant themes of the period post 1980s. The final part will involve getting a closer look at specific architects and their body of work.

Learning Outcomes :

After completing the Courses, the student will be able to :

- - Describe the thematic variations and diversity of approaches in the architectural practices of the Global South
- - Evaluation of a body of work to identify affinity to broad themes and patterns
- - To be able to define a specific and precise argument for writing a paper. To be able to use precedents as evidence for validating an argument.
- - Articulate the historical trajectories of a particular group of practices within a limited geographical area
- - To be able to present the learnings as an essay, suitably annotated with illustrations, diagrams, drawings, photographs and other visual material. The essay should use appropriate references and standard academic writing.

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	Decoding the terms; Presenting the Eurocentric influence on architectural practice; short exercise
Week 2	An overview of the architectural practices in the global south; short exercise
Week 3	An overview of the architectural practices in the global south; short exercise Preparing comparative timelines
Week 4	Discussion on thematic similarities of the body of work; Panel discussion; long question based on discussion
Week 5	Discussion on thematic similarities of the body of work; Panel discussion; long question based on discussion
Week 6	Discussion on thematic similarities of the body of work; Panel discussion; long question based on discussion
Week 7	Discussion on thematic similarities of the body of work; Panel discussion; long question based on discussion; finalisation of essay topic
Week 8	Thematic focus: Specific practices and their interpretations;
Week 9	Thematic focus: Specific practices and their interpretations; first draft submission of essay
Week 10	Thematic focus: Specific practices and their interpretations; first draft submission
Week 11	Thematic focus: Specific practices and their interpretations; final essay submission
Week 12	Thematic focus: Specific practices and their interpretations

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Code	AR3608	Course	HRW7- Architectural Theory		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

Each semester of design studios, in academics, is a fresh start with a new site, new program, and new peers! It is assumed that it all adds up in the deep recesses of the students' minds to create a cohesive understanding of, and mastery over design. Often this doesn't happen, resulting in architectural offices practicing design on a project-to-project basis. This course introduces students to the act of theorizing in architecture. Theorizing is an essential component to arrive at a design manifesto, a larger vision for one's own design practice. Hegelian dialectics is used as a technique to debate a set of opposing architecture theories. Each session picks up one aspect of architecture [e.g., role of ornament, idea of harmony, rational and the sensuous] and through three texts- thesis, antithesis, and synthesis and accompanying examples, generates a debate about the aspects. This course is broadly based on Korydon Smith's Introducing Architectural Theory. However, this course is not an introduction to the history of architectural theory. Students will be expected to critique their previous studio projects and write an essay on possible future directions for their own journey.

Learning Outcomes :

After completing the Courses, the student will be able to :

- describe the theoretical underpinnings influencing the works of Architects
- analyse ones own designs and identify positions and directions

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	Introduction. Epistemology, Ontology, Axiology.
Week 2	material and immaterial
Week 3	simplicity and complexity
Week 4	function and form
Week 5	self and the selfless
Week 6	diagram and the sensuous
Week 7	harmony in architecture
Week 8	Corbusier meets Jacobs
Week 9	place and placelessness
Week 10	true stories?
Week 11	ornament and crime
Week 12	concluding session

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Code	AR2731	Course	Generative Art		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

Defining art can be a challenge, but what elements can define art? Art is often thought to be a reflection of society, so what is the art of today and tomorrow? As we head towards the age of augmentation, can technology enhance art as well? The course revolves around these questions and finds itself at the intersection of visual arts and computational technology. It travels to the opposite extremes of the design spectrum; i.e. art and science to discover generative art. The students would get a chance to explore the chosen artwork and develop a perspective to study its aspects. This perspective would then translate into their storyline; aspects into characters. The data and the relationships established between these characters will form the raw material to create their own piece of art in the chosen medium. The entire process of decoding and encoding art would take a computational path. While artists traditionally rely on intuition and experience to create art, generative artists leverage modern processing power to invent new aesthetics – instructing programs to run within a set of artistic constraints, and guiding the process to a desired result. This method vastly reduces the exploratory phase in art and design and often leads to surprising and sophisticated new ideas.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Develop an understanding of the current art world and derive their own perspective/narrative to interpret art.
- Perceive art as a process of creation beyond its formal qualities and the notion of it being an object worth appreciation.
- Generate art through computational methods and develop basic scripting ability.
- Apply an algorithmic thinking process as an alternative to the conventional design approaches in their future projects.

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	Introduction: Generative art, its methods, mediums, and tools.
Week 2	Skill building: Understand computational workflow + Grasshopper tutorials
Week 3	Mediums: Discuss inspirations, define a perspective/narrative, and select the medium + Grasshopper tutorials
Week 4	Formulate Perspective to Representation. Develop a pseudo-code + Grasshopper tutorials
Week 5	Compose: Making a working script.
Week 6	Mid-semester review: Refining the script and producing outputs/iterations. Collate the outputs.

Week 7	Guest Lecture Output to art: Devise techniques to use the digital results and produce artwork from it. Tweaking of scripts and integration of other software in the process for graphics, animation, audio, etc.
Week 8	Speculate: Selection of iterations and propose the form, material and technique to produce art.
Week 9	Prepare: Working files for execution [analog] and generate data for compilation.
Week 10	Dry run: Trial attempt at production/printing
Week 11	Post-processing: Refining and curating the final artwork.
Week 12	Final review